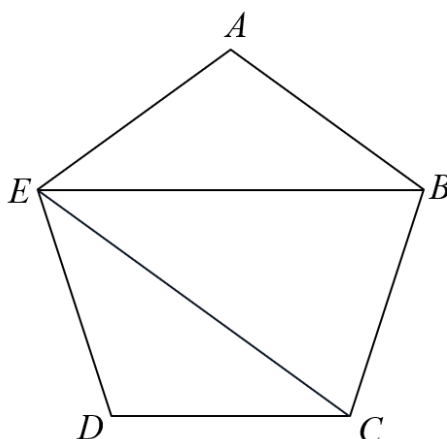


- 23 In the diagram, $ABCDE$ is a regular pentagon.



- (a) Show that BE is parallel to CD .

Answer _____

[3]

- (b) The sides AB and DC are produced to meet at X .
What type of quadrilateral is $BECX$?
Explain your answer.

Answer _____

[3]

24

A is the point $(8, -2)$ and $\overrightarrow{AB} = \begin{pmatrix} 4 \\ -3 \end{pmatrix}$.

(a) Find

(i) $|\overrightarrow{AB}|$,

Answer _____ [1]

(ii) the equation of the line AB .

Answer _____ [2]

(b) The equation of the line CD is $2y - 4x = 19$.

Find the coordinates of the point of intersection of AB and CD .

Answer (_____ , _____) [3]

-
- There are 12 boys and 13 girls in a class.
- 25** Two of the children are selected at random to participate in a forum.
- (a)** Draw a tree diagram below to show the probabilities of the possible outcomes.

[2]

- (b)** Find, as a fraction in its simplest form, the probability that
- (i)** two boys are selected,

Answer _____ [1]

- (ii)** one boy and one girl are selected.

Answer _____ [2]